



ENDOCRINE PHARMACOLOGY

Hormones and Hormone Antagonists

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- The function of the body is regulated by two major control systems:
 - ❑ The nervous system
 - ❑ The hormonal or endocrine system
 - In general, the hormonal system is concerned principally
 - ❑ With control of the different metabolic functions of the body
 - ❑ Controlling the rates of chemical reactions in the cells or the transport of substances through cell membranes or other aspects of cellular metabolism, such as growth and secretion
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Hormones include :-

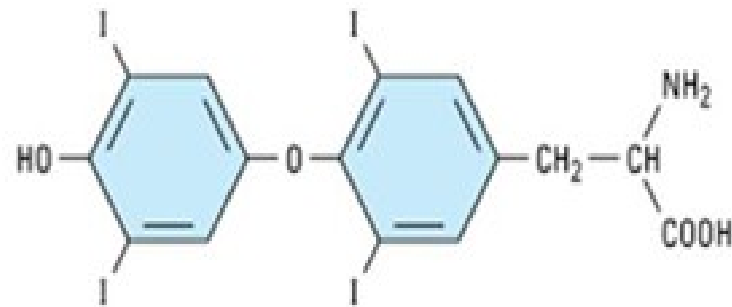
- **Anterior pituitary hormones:** *growth hormone, adrenocorticotropin, thyroid-stimulating hormone, follicle-stimulating hormone, luetinizing hormone, prolactin and melanocyte-stimulating hormone*
- **Posterior pituitary hormones:** *antidiuretic hormone (vasopressin) and oxytocin*

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- **Adrenocortical hormones:** *cortisol and aldosterone*
 - **Thyroid hormones:** *thyroxine, tri-iodo-thyronine and calcitonin*
 - **Pancreatic hormones:** *insulin and glucagons*
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- **Ovarian hormones:** *estrogen and progesterone*
 - **Testicular hormones:** *testosterone/androgen*
 - **Parathyroid hormones:** *parathormone*
 - **Placental hormones:** *chorionic gonadotropin, estrogen, progesterone and human placental lactogen*
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Thyroid and Antithyroid Drugs

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- The thyroid gland is essential for many physiological processes. It secretes three main hormones:
 - ❑ Thyroxine (T₄)
 - ❑ Triiodothyronine (T₃) and
 - ❑ Calcitonin
 - T₃ is three to fivefold more active than T₄.
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INACTIVATION

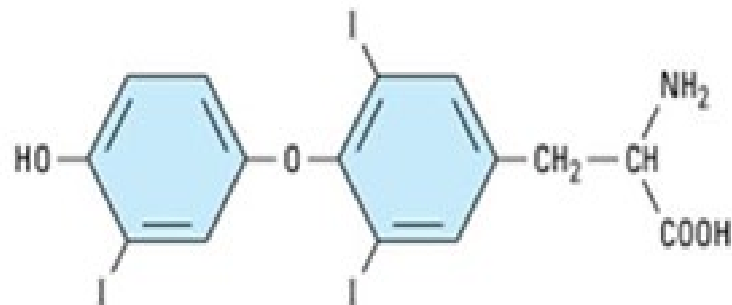
Deamination
Decarboxylation
Conjugation
(glucuronide or sulfate)

Thyroxine

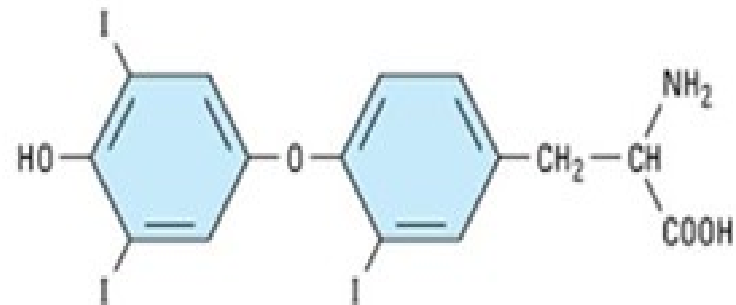
Deiodination

ACTIVATION

INACTIVATION



3,5,3'-Triiodothyronine
(T₃)



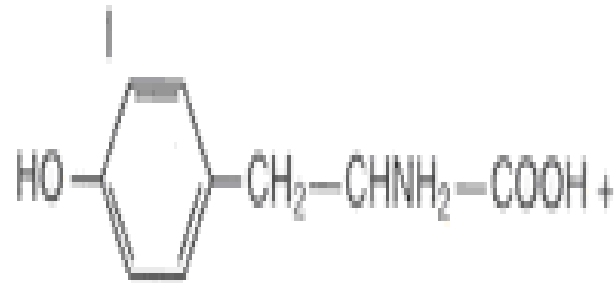
3,3',5'-Triiodothyronine
(reverse T₃)

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- T_4 and T_3 are critically important for normal growth and development and for energy metabolism
 - Calcitonin is involved in the control of plasma calcium
 - The term thyroid hormone will be used here to refer to T_3 and T_4 .
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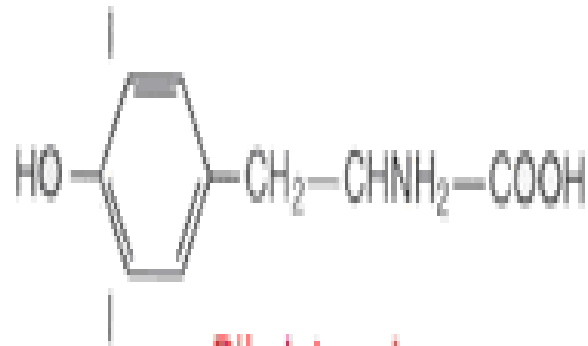
Synthesis of thyroid hormones

- The main steps in the synthesis, storage and secretion of thyroid hormone are as follows:
- Step one:-uptake of plasma iodide by the follicle cells.
 - ❑ Iodide from diet is taken by a symporter known as called sodium iodide symporter

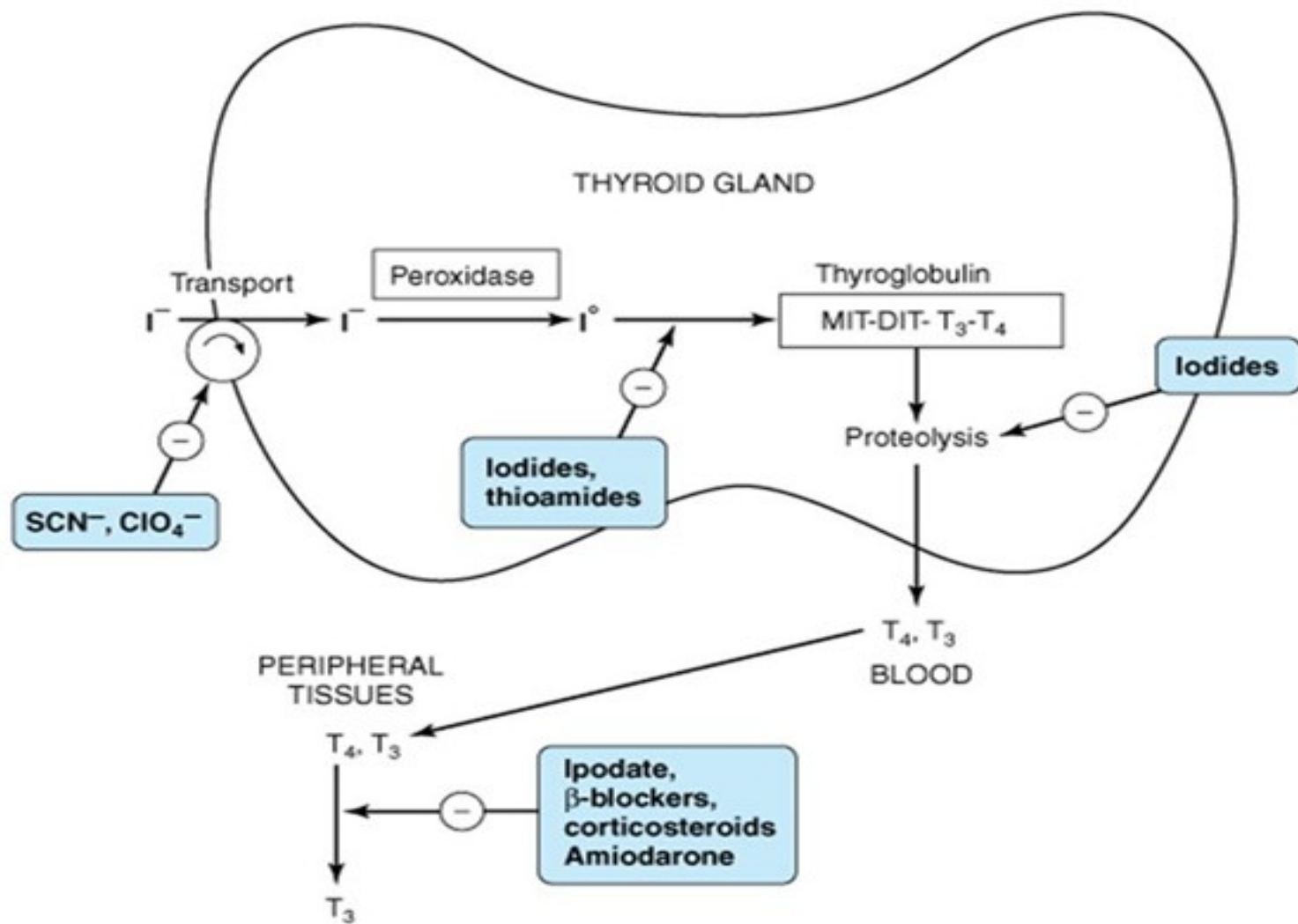
- Step two :- Oxidation of iodide and iodination of tyrosine residues in the thyroglobulin
 - ❑ Monoiodotyrosine and diiodotyrosine are formed



Monoiodotyrosine



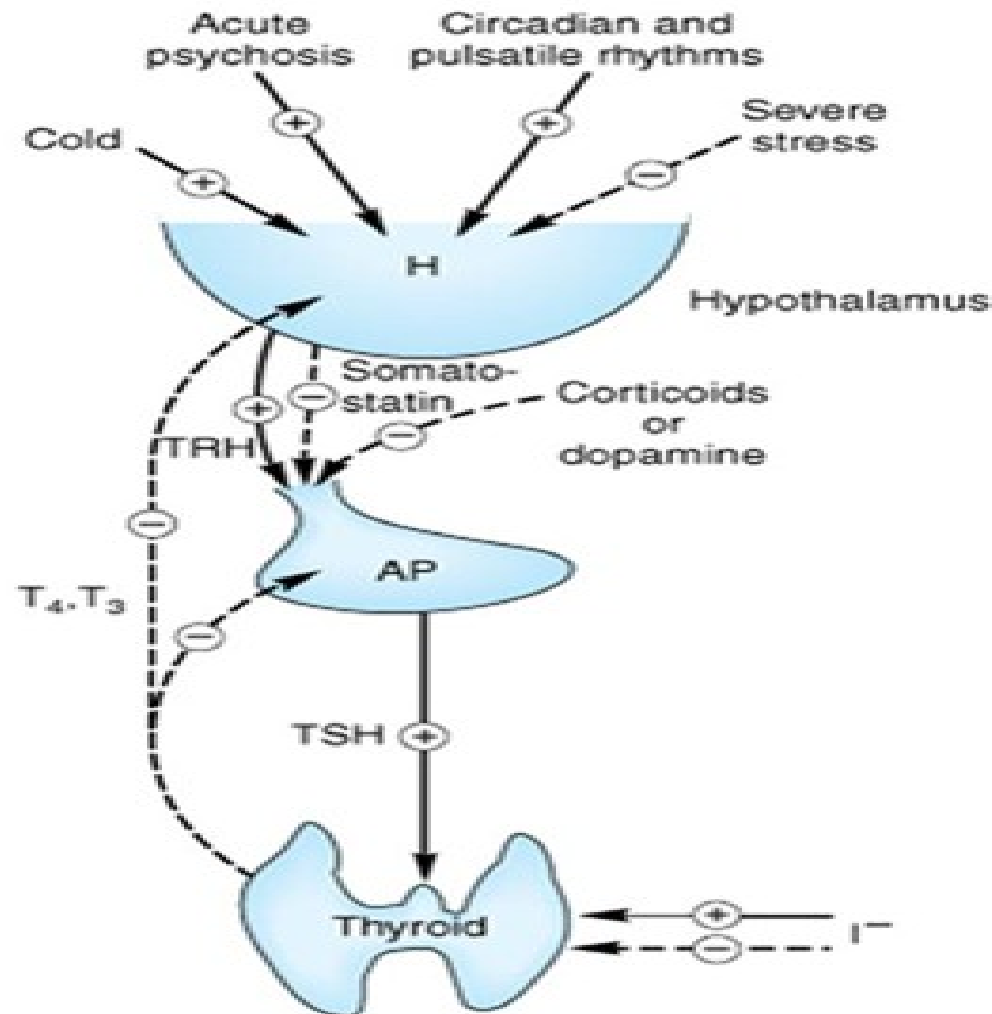
Diiodotyrosine



Actions of the thyroid hormones

- The physiological actions of the thyroid hormones fall into two categories:
 - ❑ Those affecting metabolism and
 - ❑ Those affecting growth and development

Thyroid-pituitary relationships



Effects on metabolism

- Increase metabolism of
 - Carbohydrate
 - Fat and
 - Protein
 - There is an increase in oxygen consumption and heat production which is manifested as an increase in basal metabolic rate.
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- The calorogenic action is important as part of the response to a cold environment.
 - Administration of thyroid hormone results in augmented cardiac rate and output and increased tendency to dysrhythmias such as atrial fibrillation
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Effects on growth and development

- Thyroid hormones have a critical effect on growth, partly by a direct action on cells and partly indirectly by influencing growth hormone production and potentiating its effect.
 - Also important for skeletal development.
 - They are particularly necessary for normal growth and maturation of the CNS
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➤ **Abnormalities of thyroid function**

- ❑ *Hypothyroidism*: - inadequate secretion of thyroid hormones
 - ❑ *Hyperthyroidism*: - excess secretion of thyroid hormones
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Hypothyroidism

- Clinical state that results from decreased production of thyroid hormones or decreased action at the tissue level
- Overt hypothyroidism occurs in 1.5% to 2% of women and 0.2% of men, and
- its incidence increases with age
- Classified into
 - Primary(95%)
 - Due to failure of the thyroid gland itself
 - Secondary
 - Due to hypopituitarism
 - Tertiary
 - Due to failure of the hypothalamus
 - Peripheral
 - Due to tissue insensitivity to the action of thyroid hormones

Treatment of Hypothyroidism

- The only effective treatment of hypothyroidism, unless it is due to iodine deficiency (treated with iodide) is
- To administer the thyroid hormones themselves
 - ❑ Thyroxine (levothyroxine, T_4) is the standard replacement therapy
 - ❑ Triiodothyronine (liothyronine, T_3) being reserved for the rare condition of myxoedema coma when its more rapid action is required for emergency treatment.

Levothyroxine Sodium

- Sodium salt of the naturally occurring levorotatory isomer of T4.
 - Preparation of choice for maintenance of plasma T4 and T3 concentrations
 - It is absorbed intact from the gastrointestinal tract, and its long half-life allows for convenient oncedaily administration
 - The aim is to establish euthyroidism with measured serum concentrations of T4, T3, and TSH within the normal range
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Liothyronine Sodium

- Liothyronine is generally not used for maintenance thyroid hormone replacement therapy because of its short plasma half-life and duration of action
 - The use of T3 alone is recommended only in special situations, such as in the initial therapy of myxedema and myxedema coma and the short-term suppression of TSH in patients undergoing surgery for thyroid cancer
 - Useful in 5-deiodinase deficiency
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➤ Liotrix

- ❑ is a 4:1 mixture of levothyroxine sodium and liothyronine sodium
 - ❑ used for thyroid hormone replacement therapy in hypothyroid patients
 - ❑ does not appear that liotrix offers any therapeutic advantage over levothyroxine alone
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Adverse effects of treatment with thyroid hormone

- The most common adverse effects (i.e., symptoms of hyperthyroidism) are the result of a drug overdose; they include
 - ❑ Cardiac palpitation and arrhythmias, tachycardia,
 - ❑ Weight loss, tremor, headache, insomnia, and heat intolerance
 - Symptoms subside if medication is withheld for several days
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Treatment of Hyperthyroidism

- The goal of therapy is to decrease synthesis and/or release of additional hormone.
- **Removal of part or all the thyroid gland**
 - Surgically
 - Radioiodine ^{131}I

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- Radioiodine ^{131}I :- by destruction of the gland by beta particles emitted
 - ❑ It is a first line treatment
 - ❑ It has a half-life of eight days, by two month its radioactivity has effectively disappeared
 - ❑ It is used in one single dose
 - ❑ Hypothyroidism will eventually develop, but is easily managed by replacement therapy with thyroxine.
 - ❑ It is best avoided in children and also in pregnant patients because of potential damage to the fetus.
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■ **Inhibition of thyroid hormone synthesis**

□ Thioureylenes/Thioamides

- Includes carbimazole, methimazole, propylthiouracil

➤ **Mechanism of Action**

- Not completely understood, but there is evidence that they inhibit the iodination of tyrosyl residues in thyroglobulin
- They do not inhibit secretion of stored thyroid hormone
- Propylthiouracil has the additional effect of reducing the de-iodination of T_4 to T_3 in peripheral tissues.

Mechanism contd

- Thionamide drugs interfere with peroxidase catalyzed reactions
- In the thyroid gland, they inhibit the activity of the enzyme TPO, which is required for
 - ❑ The intrathyroidal oxidation of I,
 - ❑ The incorporation of I into Tg, and
 - ❑ The coupling of iodotyrosyl residues to form thyroid hormones

Pharmacokinetics

- Thionamide drugs are well absorbed from the gastrointestinal tract.
- They have short plasmahalf-lives (propylthiouracil 1.5 hours; methimazole 6 hrs)
- Single daily dose may exert effects for greater than 24 hours.
- Thionamides undergo hepatic conjugation and are excreted in the bile and urine.
- Few glucuronide conjugates are found in the feces because they are absorbed from the gastrointestinal tract

- The thionamide drugs are used in the management
 - ❑ Hyperthyroidism
 - ❑ Thyrotoxic crisis and
 - ❑ In the preparation of patients for surgical subtotal thyroidectomy

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- Although thionamides may be used to treat hyperthyroidism during pregnancy, they should be given in minimally effective doses to avoid
 - ❑ Inducing infantile hypothyroidism and
 - ❑ Thyroid enlargement in the developing fetus
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Adverse effects

- Hypothyroidism and enlargement of the thyroid gland
- Granulocytopenia and agranulocytosis
- Rash
- Arthralgia, myalgia,
- Cholestatic jaundice,
- Lymphadenopathy
- Drug fever, psychosis, and a lupuslike syndrome

■ Iodine

- ❑ It is converted *in-vivo* to iodide which temporarily inhibits the release of thyroid hormones.
- ❑ When high doses of iodine are given to thyrotoxic patients, the symptom subsides within 1 to 2 days.
- ❑ There is inhibition of secretion of thyroid hormones and over a period of 10– 14 days, a marked reduction in vascularity of the gland, which becomes smaller and firmer.
- ❑ Iodine solution in KI (Lugol's iodine) is given orally.

➤ Mechanism of Action

- ❑ It may inhibit iodination of thyroglobulin possibly by inhibiting the H_2O_2 generation
- ❑ High concentration limits its own transport and it prevents iodination process.
- ❑ It inhibits release of the thyroid hormone.

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- KI has long been used in combination with propylthiouracil in the management of thyrotoxic crisis to rapidly inhibit thyroid hormone secretion.
 - Iodide plus a thionamide has also been used in the immediate preoperative preparation of patients about to undergo total or subtotal surgical thyroidectomy
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- Adverse reactions to iodine can be divided into
 - ❑ Intrathyroidal reactions
 - Iodine-induced thyrotoxicosis (Jod-Basedow's phenomenon)
 - Iodide goiter or hypothyroidism
 - ❑ Extrathyroidal reactions
 - Rash ,drug fever; sialadenitis (inflammation of the salivary glands)
 - Conjunctivitis and rhinitis
 - Vasculitis ; and a leukemoid eosinophilic granulocytosis
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Potassium Perchlorate

- The perchlorate ion of potassium perchlorate, KClO_4 , is a competitive inhibitor of thyroidal I transport via the Sodium Iodide Symporter (NIS)
 - This drug can cause fatal aplastic anemia and gastric ulcers and is now rarely used
 - effective in treating iodine-induced hyperthyroidism
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➤ Others

- ❑ Beta adrenoceptor antagonists e.g. propranolol
 - For blunting the widespread sympathetic stimulation.
- ❑ Guanethidine: a noradrenergic blocking agent
 - To ameliorate the exophthalmia, eye drops, act by relaxing the sympathetically innervated smooth muscle that causes eyelid retraction.
- ❑ Glucocorticoids: e.g. prednisolone for the exophthalmia of Graves disease

Thyroid Treatment: Potential Drug Interactions

- Drugs that reduce thyroid hormone production
 - ❑ Lithium
 - ❑ Iodine-containing medications
 - ❑ Amiodarone (Cordarone)
- Drugs that reduce thyroid hormone absorption
 - ❑ Sucralfate (Carafate)
 - ❑ Ferrous sulfate (Slow Fe)
 - ❑ Cholestyramine (Questran)
 - ❑ Colestipol (Colestid)
 - ❑ Aluminum-containing antacids
 - ❑ Calcium products

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- Drugs that increase metabolism of thyroxine
 - ❑ Rifampin (Rifadin)
 - ❑ Phenobarbital
 - ❑ Carbamazepine (Tegretol)
 - ❑ Warfarin (Coumadin)
 - ❑ Oral hypoglycemic agents
 - Drugs that displace thyroid hormone from protein binding
 - ❑ Furosemide (Lasix)
 - ❑ Mefenamic acid
 - ❑ Salicylate
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Pancreatic Hormones and Antidiabetic Drugs

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- The pancreas is both an
 - ❑ **Endocrine gland** that produces the peptide hormones *insulin, glucagon, somatostatin* and *pancreatic polypeptide*, and
 - ❑ **Exocrine gland** that produces *digestive enzymes*
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➤ Insulin

- ❑ It controls the metabolism of carbohydrate, fat and protein and normally determines blood glucose concentration

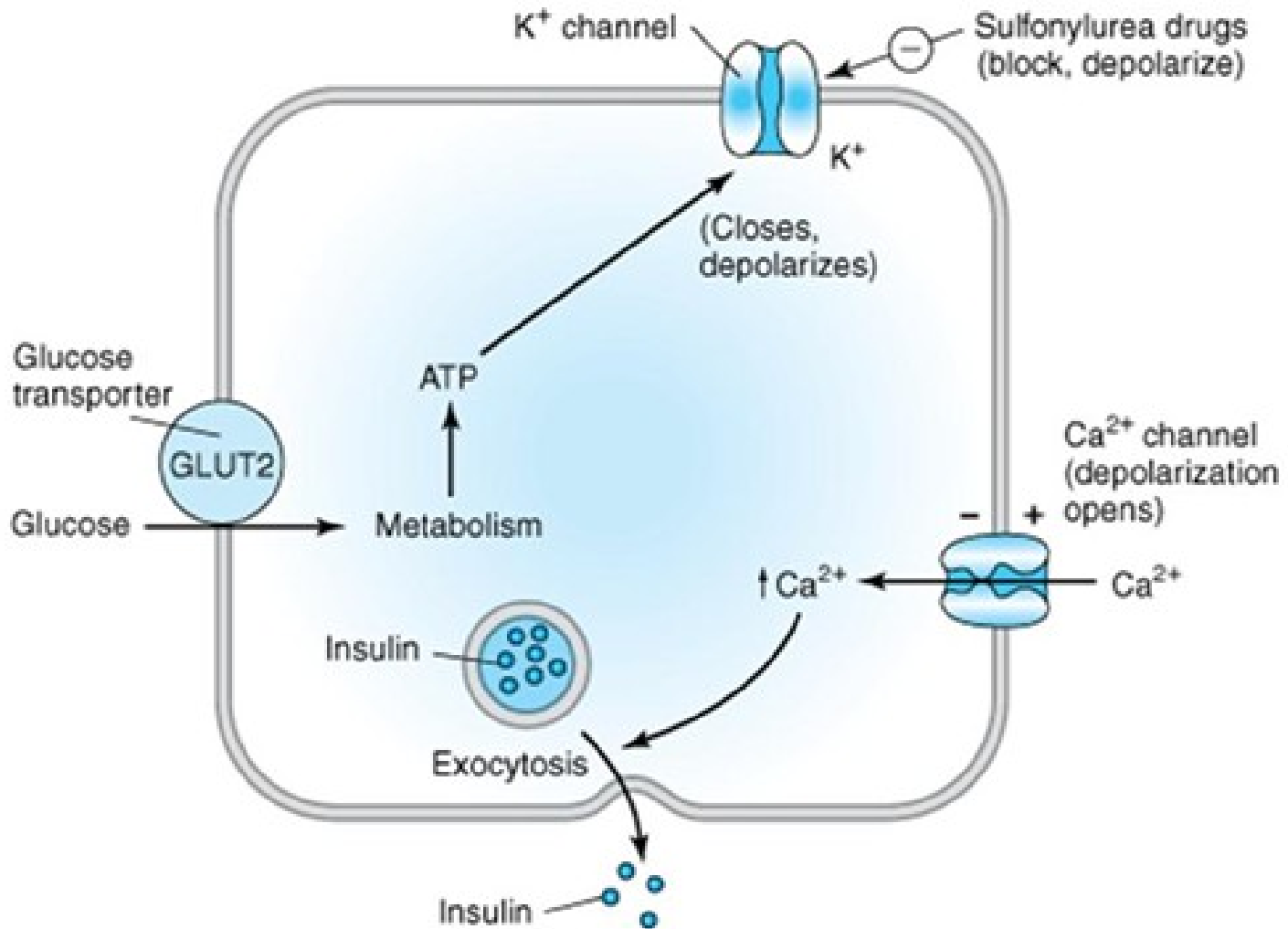
➤ Glucagon

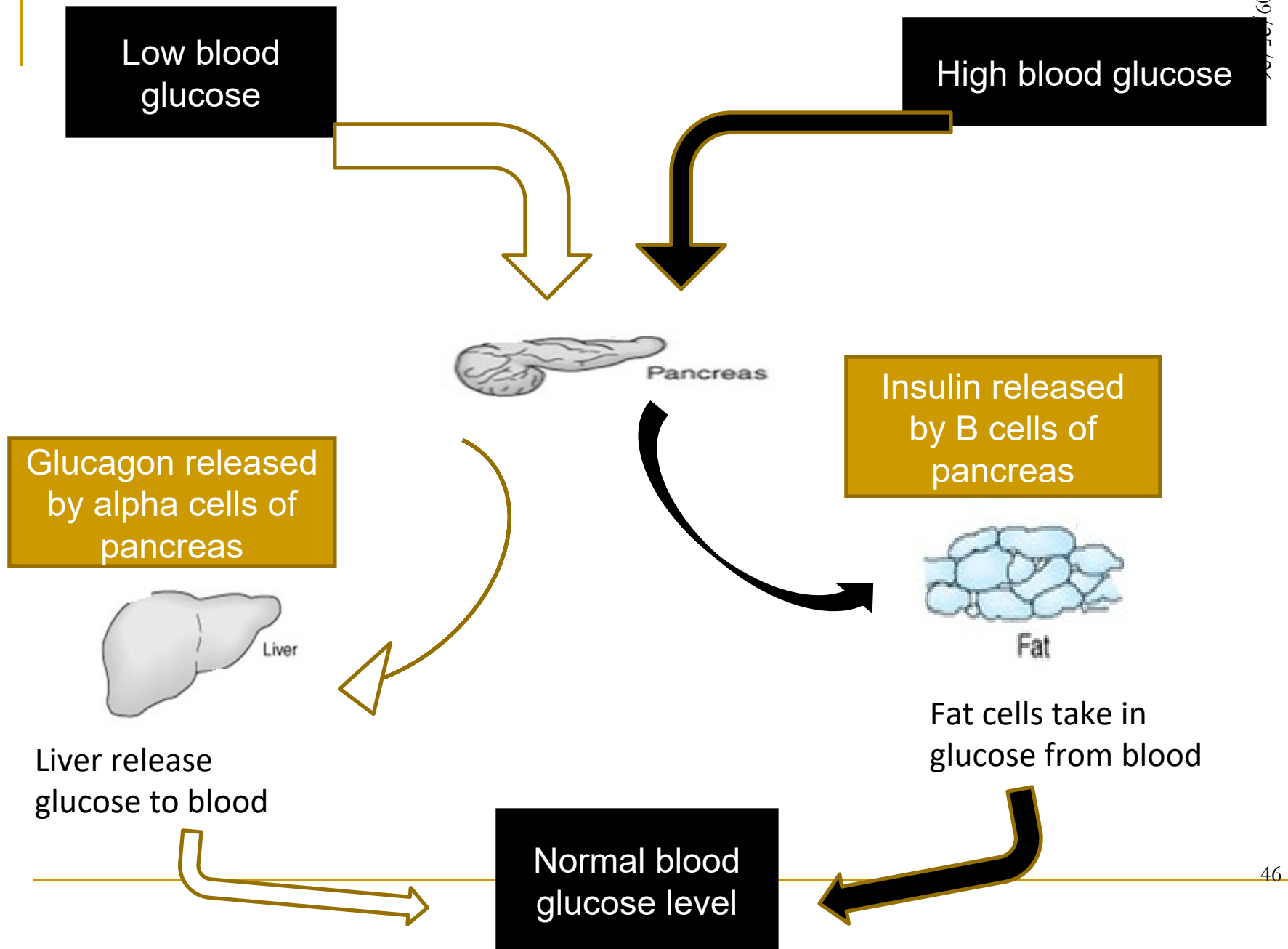
- ❑ It increases blood glucose level

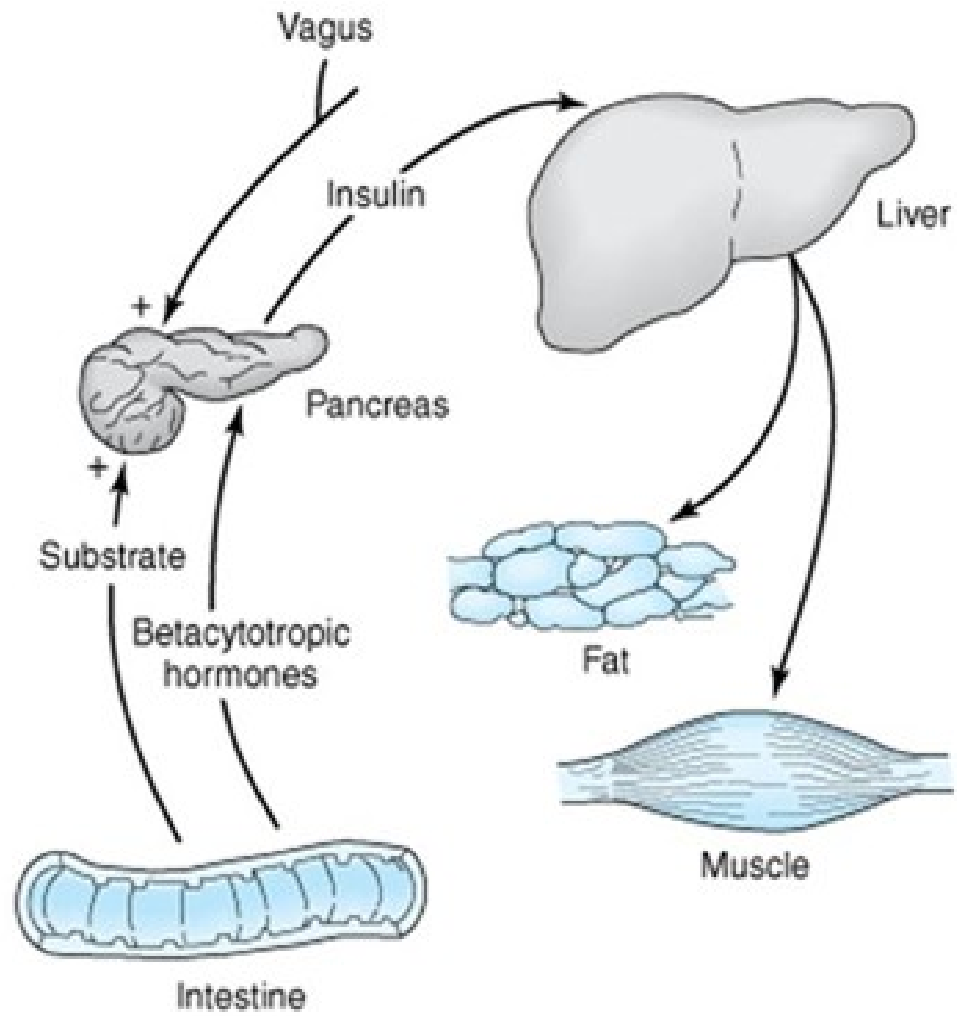
➤ Somatostatin

- ❑ It has an indirect paracrine role in that it inhibits secretion of insulin and glucagon.

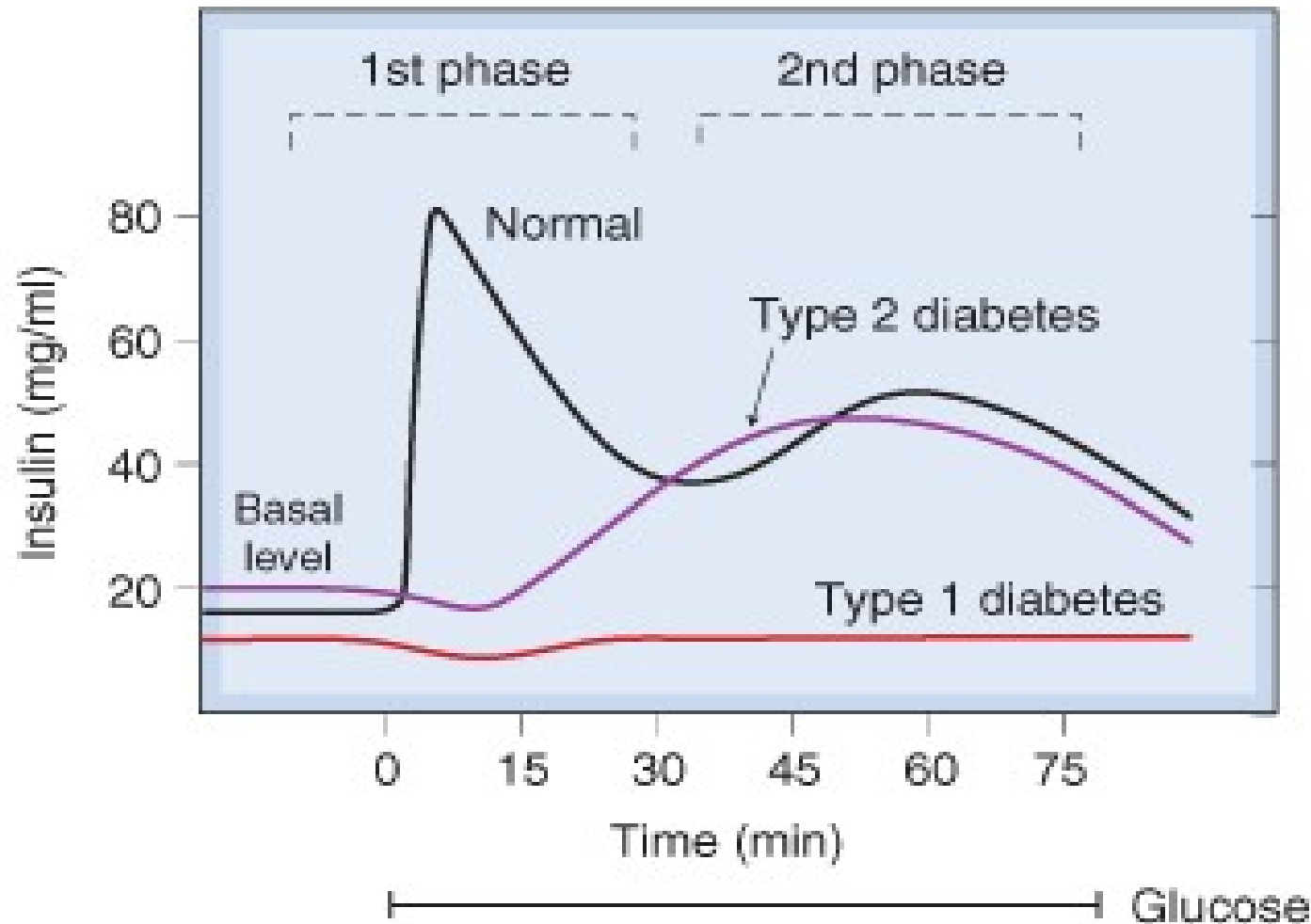
Secretion of insulin from B-cells of the pancreas





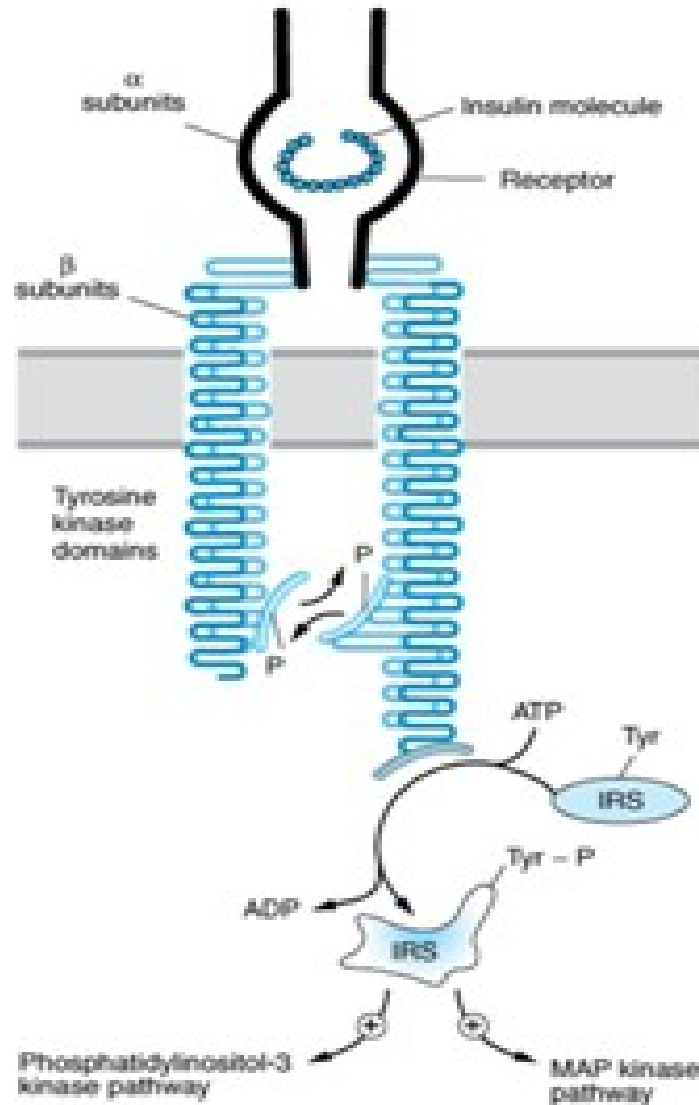


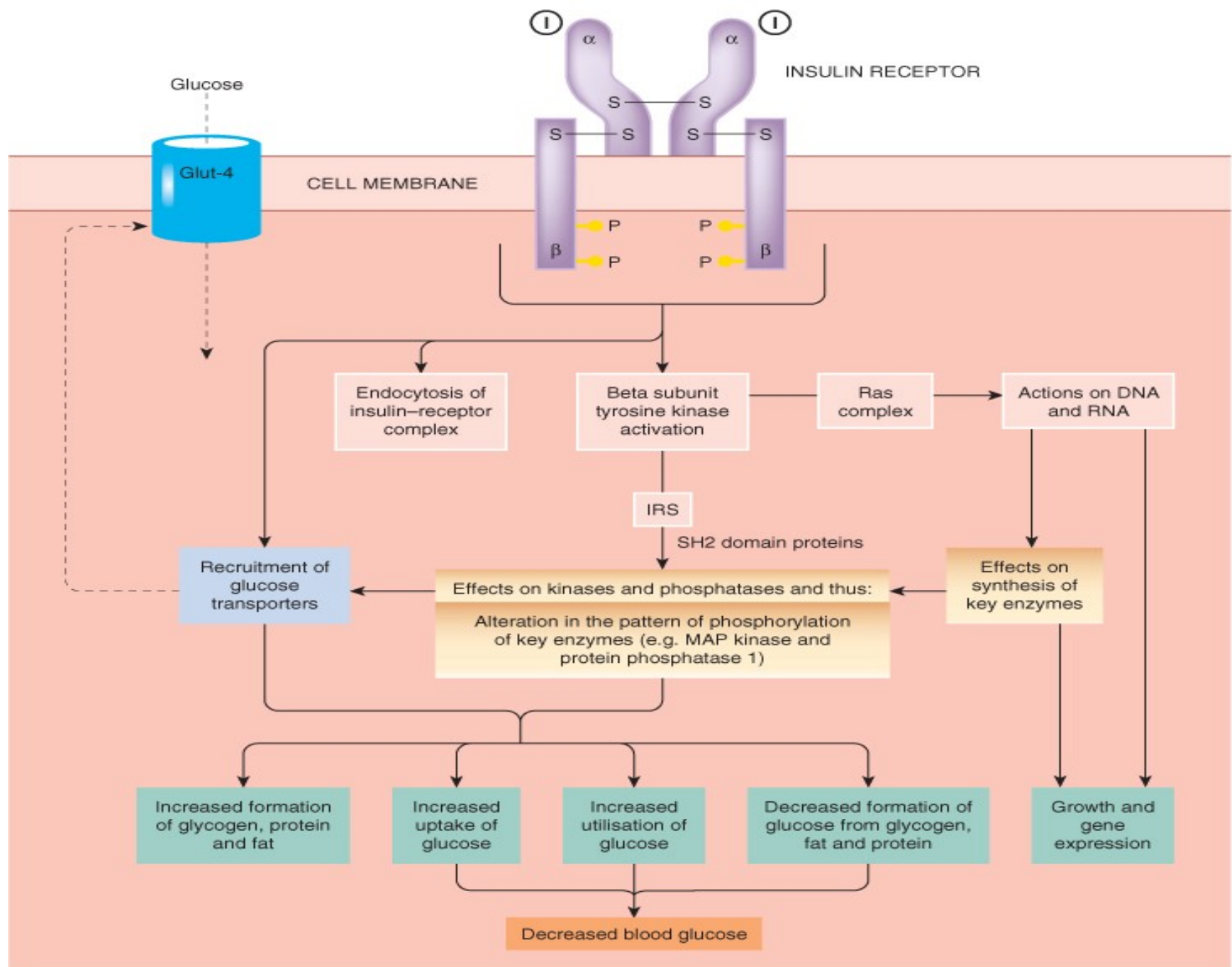
Release of insulin: occurs in two phases



Mechanism of action of insulin

09/25/22





Effects of insulin on carbohydrate, fat and protein metabolism

Type of metabolism	Liver cells	Fat cells	Muscle
Carbohydrate metabolism	↓Gluconeogenesis	↑Glucose uptake	↑Glucose uptake
	↓Glycogenolysis	↑Glycerol synthesis	↑Glycolysis
	↑Glycolysis		↑Glycogenesis
	↑Glycogenesis		
Fat metabolism	↑Lipogenesis	↑Synthesis of triglycerides	
	↓Lipolysis	↑Fatty acid synthesis	
		↓Lipolysis	
Protein metabolism	↓Protein breakdown		↑Amino acid uptake
			↑Protein synthesis

Diabetes Mellitus

- Diabetes is disease characterized by an elevation of blood glucose caused by a *relative* or *absolute* deficiency of insulin.
 - There are two main forms of Diabetes mellitus.
 - ❑ Type I diabetes
 - ❑ Type II diabetes
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Type I diabetes

- It is also known as insulin-dependent diabetes mellitus (IDDM) or juvenile-onset diabetes mellitus.
- It accounts for 10 - 20% of diagnosed diabetics.
- there is an *absolute deficiency* of insulin resulting from autoimmune destruction of β -cells

- Such patients are usually young (children or adolescents) and not obese when they first develop symptoms, but it can also occur among adults.
- Loss of β -cell function may be due to invasion by viruses, the action of chemical toxins, or usually through the actions of autoimmune antibodies directed against β -cell
- *Treatment* : exogenous insulin
 - Unless insulin treatment is provided the patient will die with diabetic ketoacidosis

Type II diabetes

- It is also known as Non-insulin-Dependent Diabetes Mellitus (NIDDM) or Maturity - Onset Diabetes Mellitus
- Most diabetics are in this category (80 -90% of diagnosed diabetics)
- Genetic factors rather than viruses or autoimmune antibodies are apparently casual (very strong genetic predisposition).

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- Occur predominantly in adults but occasionally in adolescents
 - The pancreas retains some B-cell function, resulting invariable insulin levels that are insufficient to maintain glucose homeostasis.
 - But circulating endogenous insulin is sufficient to prevent ketoacidosis
 - Obesity, which generally results in impaired insulin action, is a common risk factor for this type of diabetes, and most patients with type II diabetes are obese
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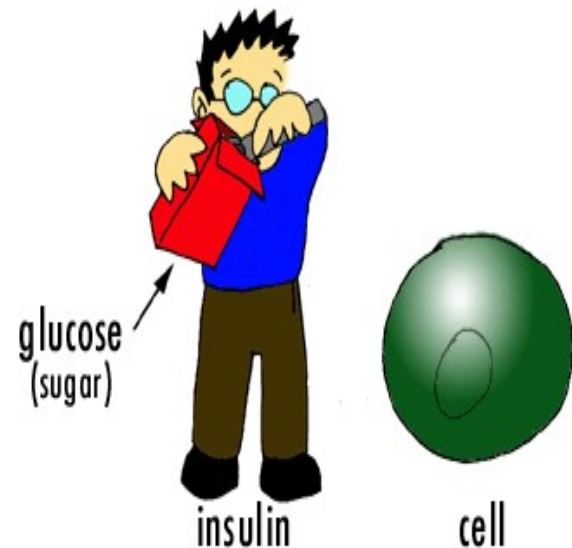
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- Tissue insensitivity to insulin (insulin resistance) is the cause of hyperglycemia
 - There is an accompanying deficiency of the pancreatic β cell's response to glucose.
 - In some cases, insulin resistance is due to a decreases number or mutations of insulin receptors. However, an as yet undefined defect in the events that occur after insulin binds to its receptor is believed to account for resistance in most patients.
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➤ Treatment

- Initially weight reduction, exercise and dietary modification; which decrease insulin resistance and correct the hyperglycemia.
 - Oral hypoglycemic drugs (sulfonylurea)
 - Insulin therapy may be required to achieve satisfactory serum glucose level even though it is not needed to prevent ketoacidosis in patient with type II diabetes
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Drugs for the Treatment of Diabetes Mellitus

- Insulin
- Oral hypoglycemic agents

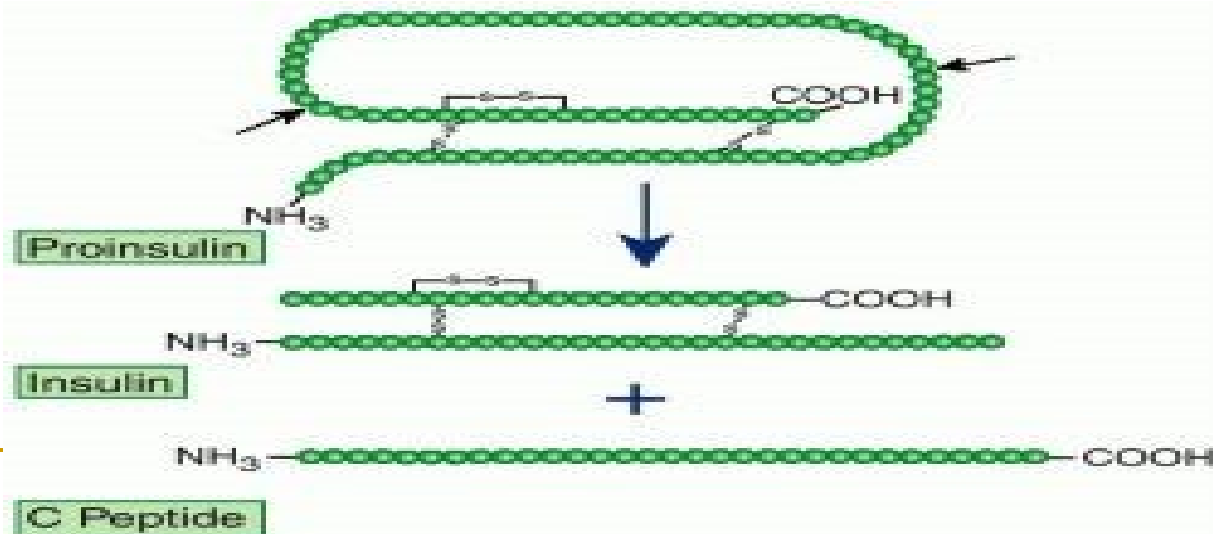


Insulin

- It is the main hormone controlling intermediary metabolism, having action on liver, muscle and fat.
 - Its overall effect is to conserve fuel by facilitating the uptake, utilization and storage of glucose, amino acids and fats after meal. Acutely, it reduces blood sugar.
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➤ Sources

- ❑ Insulin is isolated from beef and pork pancreas
- ❑ However human insulin is produced by a special strain of *Escherichia coli* that has been genetically altered to contain the gene for human insulin.
- ❑ Pork insulin is closest in structure to human insulin, differing by only one amino acid



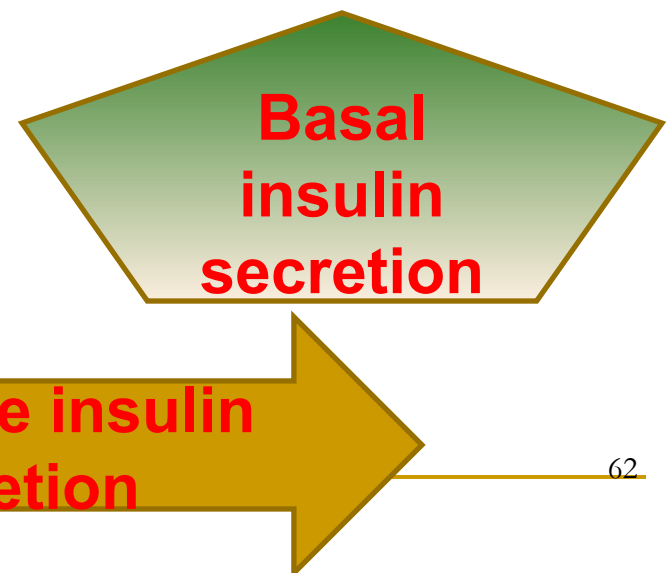
➤ Administration

- ❑ Because insulin is a protein, it is degraded in the GIT if taken orally. It is therefore generally administered parenterally usually subcutaneously, but IV or occasionally IM in hyperglycemic emergency

■ Goal of insulin administration To mimic

❑ Basal insulin secretion

- Overnight
- Fasting
- Between meals
- Prandial



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- Insulin preparations differs in
 - ❑ Recombinant technique employed
 - ❑ a.a sequence
 - ❑ Stability
 - ❑ Solubility
 - ❑ Onset of action
 - ❑ Duration of action
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➤ Insulin Preparations

- ❑ Three type of insulin preparations are available

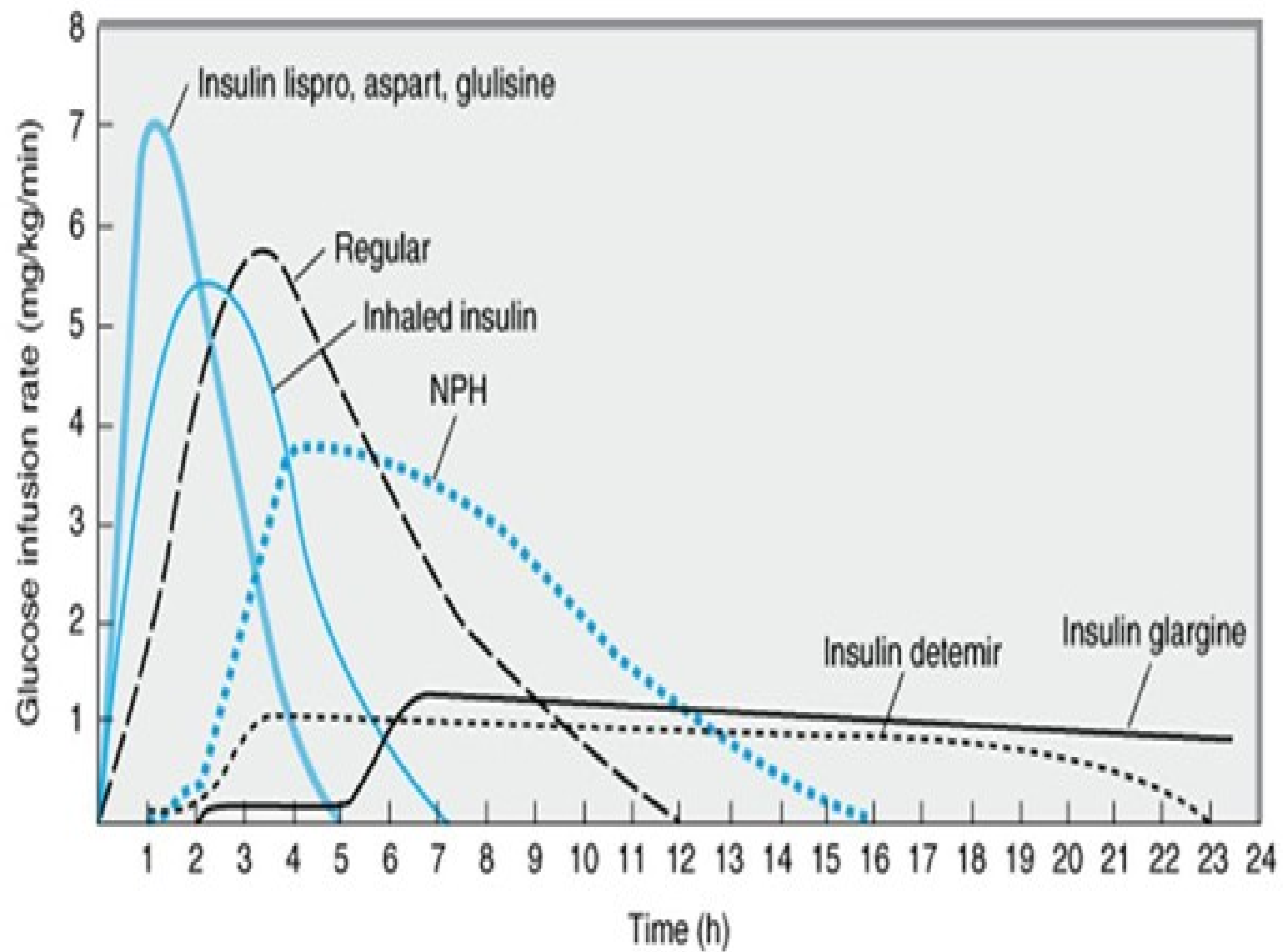
Type	Appearance	Onset (hr)	Duration (hr)
i. Rapid ▪Regular soluble (Crystalline) ▪Insulin Lispro (ultra short)	Clear Clear	0.5 -0.7 0.25	5-8 2-5
ii). Intermediate ▪Neutral protamin Hagedorn (NPH) or isophane ▪Lente (semilente insulin)	Cloudy Cloudy	1-2 1-2	18-24 18-24
iii). Slow ▪Ultralente ▪Protamine zinc ▪Glargine	Cloudy Cloudy Clear	4 -6 4 -6 2-5	20 -36 24 -36 18 -24

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- Regular insulin is short acting soluble crystalline zinc insulin. It is usually given subcutaneously (or IV in emergency) and lower blood sugar within minutes.
 - It is the only insulin preparation suitable for IV administration
 - Insulin lispro: Produced by recombinant technology wherein the amino acids on B28 and B29 (proline – lysine, respectively) have been reversed in position.....Lispro
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- NPH (neutral protamine hagedorn, or isophane) insulin
 - ❑ Prepared by treating insulin with appropriate amounts of protamine-hagedorn so as to delay absorption and the onset of action
.....neutral PH.
 - Protamine forms zinc precipitate at injection site....
Delayed and sustained release
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- Lente insulin (Insulin zinc suspension)
 - ❑ Suspension of insulin in acetate buffer at neutral PH
 - ❑ Physical state and crystal size influences the rate of absorption from injection site
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- Ultralente insulin
 - Crystalline zinc insulin particles
 - Slow absorption
 - Provide basal level of insulin
 - Insulin glargine
 - “Peakless” (having a broad plasma concentration plateau)
 - Ultra-long-acting insulin analog in which Asparagine at A 21 is replaced by Glycine and 2 Arginine residues added to the C-terminal of the B-chain
 - Should not be mixed with insulin preparations
 - Injected once daily basis
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➤ Insulin dose regimen

- ❑ Patients with type I DM can't rely on a single daily injection of insulin
- ❑ Intermediate and long acting insulins take longer time to reach peak therapeutic effect
- ❑ Thus, necessitate supplements of short and rapid acting insulins right before meals
- ❑ For convenience, these are often mixed together in the same syringe before injection

➤ Physiologic insulin dosing

- ❑ Administration of basal insulin **PLUS**
- ❑ Bolus short/rapid acting insulin **TO**
- ❑ Mimic physiologic secretion from pancreas

- It entails ----->>>>>>>>>

- ❑ More frequent injections,
- ❑ Greater reliance on short acting insulin,
- ❑ More frequent plasma glucose monitoring

- A representative physiologic dosing regimen

- ❑ Intermediate/long acting insulin as BASAL supplement
- ❑ Rapid acting insulin as a bolus

➤ Commonly used regimen for type I DM

A/. Twice daily injection of an intermediate insulin (NPH/Lente) mixed with short acting insulin (Lispro/Aspart)

➤ Drawbacks :

- Doesn't produce a near-normal glycemic control
- Rigid schedulerepast

B/. Multiple component insulin regimen/ Multiple dose insulin (MDI)

- ❑ Long acting basal insulin supplement
- ❑ Pre-prandial short acting insulin
- ❑ Dose adjusted based on,.,.,.,.,.,.,.,.
 - SMBG results
 - Anticipated food intake
 - Physical activity
- Offers better more flexible dosing and ease life style
Likely to achieve euglycemia
- Basal insulin glargine + pre-prandial Lispro/Aspart or
- 2 equal doses of Ultralente (breakfast & evening) + pre-prandial Lispro/Aspart
- Continuous subcutaneous insulin infusion (CSII)

Side effects

- Hypoglycemia- main undesirable effect
 - Treatment of hypoglycemia is to take a sweet drink or snack or if the patient is unconscious IV glucose (50% W/V soln) or IM glucagon.
- Allergy
- lipodystrophy at injection site i.e. fat accumulation may occur at the site of injection.
- Rebound hyperglycemia ('Somogyi effect') can follow after excessive insulin administration

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- Rebound hyperglycemia results from the release of the insulin-opposing or counter-regulatory hormones in response to insulin-induced hypoglycemia.
 - This can cause hyperglycemia before breakfast following an unrecognized hypoglycemic attack during sleep in the early hours of the morning.
 - It is essential to recognize this possibility to avoid the mistake of increasing (rather than reducing) the dose of insulin in this situation
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Oral Hypoglycemic Agents

- Are useful in the treatment of patients who have NIDDM but cannot be managed by diet alone.
- The patient most likely to respond well to oral hypoglycemic agents is one who develops diabetes after age 40 and has had diabetes less than 5 years.
- Patient with long-standing disease may require a combination of a hypoglycemic drug and insulin

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- Oral hypoglycemic agents includes

- **Sulfonylureas**

1. First generation

- Tolbutamide, Acetohexamide, Tolazamide, Chlorpropamide

2. Second generation

- Glyburide (glibenclamide), Glimepiride, Glipizide
 - Are more potent
 - Have fewer ADRs and DDIs
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- Advantages of 2nd gen' SUs over 1st gen' SUs
 - ❑ Higher specificity and affinity for the sulfonylurea receptor
 - ❑ More predictable pharmacokinetics in terms of time of onset and duration of action, and they have
 - ❑ Fewer side effects
 - Mechanisms of action
 - ❑ They stimulate insulin release from the β -cells of the pancreas
 - ❑ They reduce serum glucagon levels
 - ❑ They Increase binding of insulin to target tissues and receptors.
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➤ Pharmacokinetics

- ❑ Well absorbed after oral administration, and most reach peak plasma concentrations within 2-4 hours
- ❑ Metabolized by the liver and are excreted by the liver or kidney
 - Contraindicated in patients with hepatic or renal insufficiency (Because their accumulation in such patients results in hypoglycemia)
- ❑ The sulfonylureas traverse the placenta and can deplete insulin from the fetal pancreas.
 - Therefore the NIDDM pregnant women should be treated with insulin alone.
 - Also contraindicated in breast feeding

➤ Side Effects

- ❑ Hypoglycemia
- ❑ Stimulate appetite => weight gain
- ❑ Cardiovascular deaths (Because they non-selectively block K^+ channels found in the heart or blood vessels)

➤ Contraindications

- ❑ Sulphonyl Ureas are contraindicated in
 1. Obese diabetic patients
 2. Patients with hepatic or renal insufficiency
 3. Obese pregnant women
 4. Alcoholic patients (because Sulphonyl Ureas inhibit alcohol metabolism)

➤ Drug interactions

- Several drugs augment the hypoglycaemic effect of the sulfonylureas
 - ❑ Non-steroidal anti-inflammatory drugs
 - ❑ Coumarins
 - ❑ some uricosuric drugs (e.g. sulfinpyrazone),
 - ❑ Alcohol
 - ❑ Monoamine oxidase inhibitors,
 - ❑ Some antibacterial drugs (including sulfonamides, trimethoprim and chloramphenicol)
 - ❑ Some imidazole antifungal drugs

- The probable basis of most of these interactions is
 - ❑ competition for metabolizing enzymes,
 - ❑ interference with plasma protein binding
 - ❑ excretion may play some part
- Agents that decrease the action of sulfonylureas on blood glucose include
 - ❑ High doses of thiazide diuretics and
 - ❑ Corticosteroids
- Clinical use
 - Sulfonylureas require functional B cells, so they are useful in the early stages of type 2 diabetes. They can be combined with metformin or with thiazolidinediones.

➤ Tolbutamide

- ❑ The *safest sulfonylurea for elderly diabetics*, because of its short half-life
- ❑ Elimination half-life of 4–5 hours

➤ Acetohexamide

- ❑ the only sulfonylurea with uricosuric activity, an action that may be of benefit in diabetic patients who also have gout.

➤ Chlorpropamide

- ❑ Half life of 32 hours and is slowly metabolized in the liver to products that retain some biologic activity
- ❑ It is contraindicated in patients with hepatic, renal insufficiency, and in elderly patients
- ❑ Prolonged hypoglycemic reactions are more common in elderly patients
- ❑ Interaction with Dx that affect and/or are affected by hepatic metabolism

➤ **Biguanides**

- ❑ They includes :-Metformin, Phenformin, Butformin

➤ **Mechanisms of action**

- ❑ Complex and incompletely understood
- ❑ They decrease glucose production by liver
- ❑ They increase glucose uptake by skeletal muscles
- ❑ They lower plasma lipids (LDL)

➤ Advantage of Biguanides over Sulphonyl Ureas:

- ❑ Biguanides suppress appetite (therefore they can be safely be given for Obesic diabetic patients)
- ❑ They can be safely used in pregnant

➤ Side effects

- GI disturbance
- Rarely, potentially fatal lactic acidosis
- Long-term use may interfere with B₁₂ absorption

➤ Contraindications

- Renal and hepatic insufficiency

➤ **Alpha glucosidase inhibitor**

- ❑ Only monosaccharides are absorbed from the intestines
- ❑ Complex starches & bigger sugars should be digested by intestinal enzymes
 - α - glucosidases (sucrase, maltase, dextranase, glycoamylase)
 - β –glucosidases (lactase ,...)
- ❑ Inhibition of these enzymes decrease digestion and absorption of sugars ... ↓postprandial hyperglycemia
- ❑ Drugs :- Acarbose, Miglitol

➤ Acarbose

- ❑ Orally active drug for treatment of patients with NIDDM and as a possible adjunct to insulin for those with IDDM.
- ❑ It inhibits alpha-glucosidase in the intestinal brush border and thus decreases the absorption of starch and disaccharides. Consequently the postprandial rise of blood glucose is blunted.
- ❑ Does not cause hypoglycemia
- ❑ Devoid of inhibitory effect on lactase Would not be expected to cause lactate intolerance
- ❑ *Insulin sparing activity*

➤ Uses_

- ❑ Mono Tx for type II
- ❑ Adjunctive Tx with other OHGA (SU, Metformin, etc)
- ❑ Diabetes prevention

➤ Side Effects

- ❑ Flatulence
- ❑ Diarrhea
- ❑ Abdominal cramping

➤ Meglitol

- ❑ Similar mzm of action with acarbose except that it blocks β -glucosidases in addition , , , thus inhibit digestion of lactose , , , lactate intolerance

➤ Others

➤ Insulin Secretagogues

- ❑ Meglitinides:- Repaglinide
 - ❑ The meglitinides are a new class of insulin secretagogues
 - ❑ These drugs modulate β cell insulin release by regulating potassium efflux through the potassium channels.
 - ❑ Are helpful in patients with a known allergy to sulfa drugs
-

-
- ❑ Nateglinide binds with a higher affinity than does repaglinide and has a faster onset of action and a shorter duration of action.
 - ❑ metabolized by the liver and excreted by the kidney.
 - ❑ Hypoglycemia is the common SE with meglitinides
-

➤ Thiazolidinediones

- ❑ Includes: rosiglitazone, pioglitazone, troglitazone
- ❑ Are recently introduced classes of oral antidiabetic drugs that enhance target tissue insulin sensitivity.
- ❑ Can cause weight gain and oedema
- ❑ Are peroxisome proliferator-activated receptor- γ (a nuclear receptor) agonists
- ❑ PPAR receptors are predominantly expressed in
 - Adipose tissue and to a lesser extent in
 - Muscle
 - Liver and other tissues

-
- For DM, a major site of Tzd action is the adipose tissue,
 - ❑ Glucose uptake and utilization
 - ❑ Modulation of synthesis of lipid hormones or cytokines and other proteins involved in energy regulation.
 - The ultimate benefit of TZDs in DM is reduction of insulin resistance
-

-
- Substantial benefit for patients with significant insulin resistance as in patients
 - Who do not respond to other oral therapies
 - Who require excessive amounts of insulin [100 units/day]
 - TZDs are found to reduce incidence of new cases of Type
 - Tzds have a slow onset and offset of activity over weeks or even months
-

-
- An adverse effect common to Tzds is
 - Fluid retention,
 - Peripheral edema and
 - Dose-related weight gain
 - Should not be used in patients with CHF and liver diseases
 - ❑ Structural and histopathological changes in cardiac tissue;
-

Oxytocics

- They are group of drugs that cause contraction of the uterus
- ***Oxytocin***
 - It is a peptide hormone secreted by the posterior pituitary
- **Actions**
 - ❑ Contracts the uterus
 - ❑ Cause contraction of myoepithelial cells surrounding mammary alveoli, which leads milk ejection
 - ❑ It has weak antidiuretic and pressor activity

➤ Therapeutic uses

- ❑ To induce labor and augment dysfunction labor
- ❑ For control of postpartum hemorrhage
- ❑ Impaired milk ejection

➤ Side effects

- ❑ Oxytocin may cause over stimulation and leads to rupture of the uterus in the presence of cephalo-pelvic disproportion
- ❑ When given intravenously may cause water retention leading to water intoxication.

➤ *Ergometrine*

- ❑ It is one of the ergot alkaloids with the ability to cause contraction of the uterine smooth muscle.
- ❑ It causes sustained uterine contraction.
- ❑ It is completely absorbed after subcutaneous and intravenous administration,
- ❑ Metabolized in the liver and eliminated in the urine .liver damage enhances the toxicity of ergot alkaloid.

❑ *Use:-*

- After delivery of placenta if bleeding is severe
- Prevent postpartum bleeding

❑ *Adverse effect:*

- Nausea,
- vomiting but serious toxic effects are rare

➤ ***Prostaglandins***

- ❑ stimulate both the tone and amplitude of the uterine contraction
 - ❑ Adverse reaction: nausea , vomiting, headache ,diarrhea, fever ,etc
-

The gonadal hormones and inhibitors

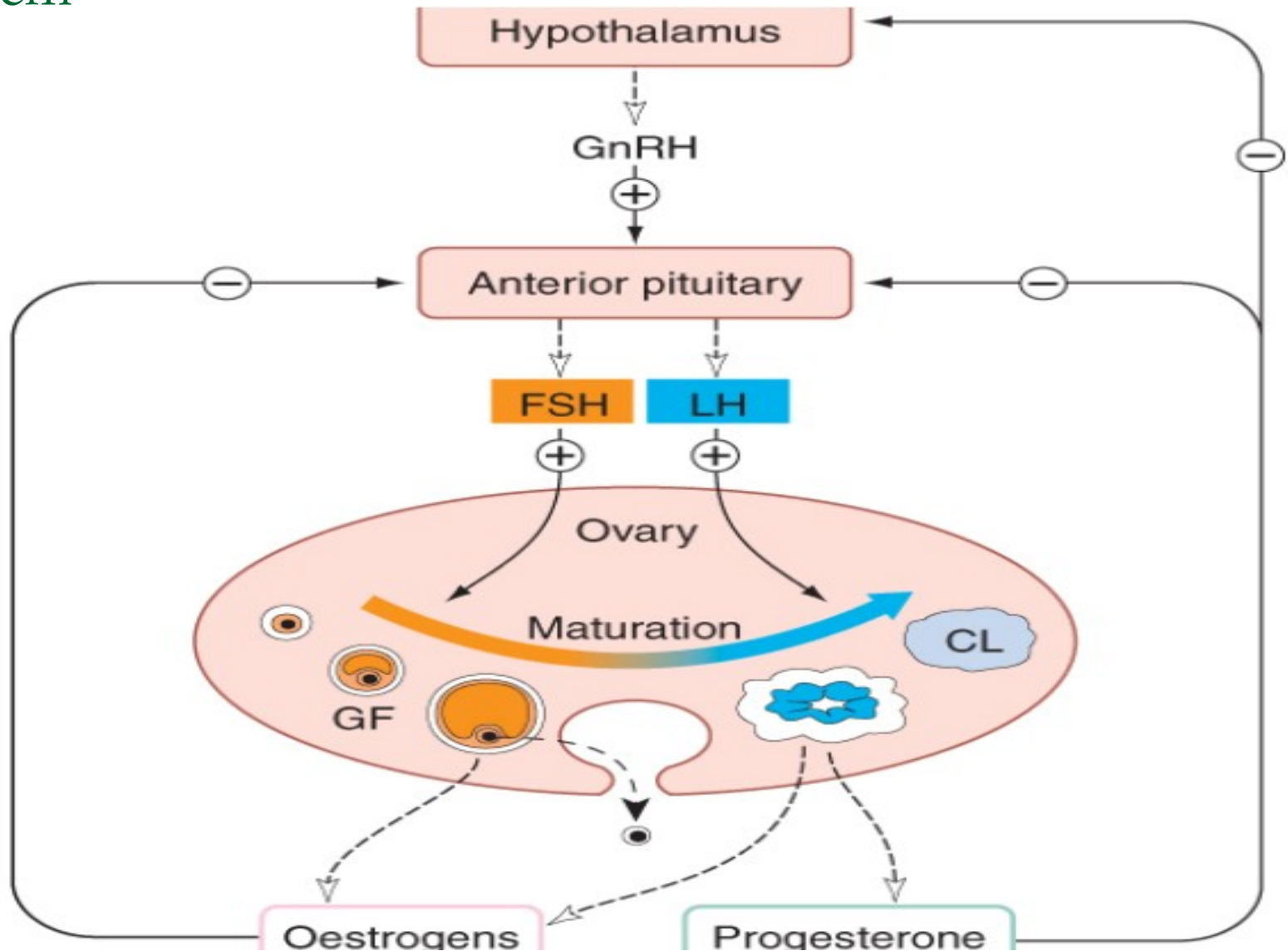
- ❖ Estrogens and progestins
- ❖ Androgens

Estrogens

➤ Includes

- ❑ Steroidal natural : estradiol, estrone, estriol
- ❑ Steroidal synthetic : ethynyl
estradiol, mestranol, quinestrol, dienestrol, benzestrol, hexestrol, methestrol
- ❑ Non steroidal synthetic : diethylstilbestrol, chlorotrianisene, methallenestril

Neurohormonal control of the female reproductive system



➤ Pharmacological effects

- ❑ Female sexual maturation
 - ❑ Endometrial effect: continuous exposure to estrogens for prolonged periods leads to hyperplasia of the endometrium that is usually associated with abnormal bleeding patterns
 - ❑ Metabolic and cardiovascular effects
 - It decreases the rate of resorption of bone
 - Causes some degree of retention of salt and water
 - The conc of serum triglycerides and of HDL is raised, whereas that of LDL is decrease
 - It increase the coagulability of the blood
-

■ Therapeutic uses

- ❑ Replacement therapy as in hypo-ovarian conditions
- ❑ Treatment of menopausal symptoms or for postmenopausal replacement therapy
- ❑ Contraception
- ❑ Vaginitis
- ❑ Prostate cancer
- ❑ Combined with progestins for intractable dysmenorrhea, hirsutism and amenorrhea

➤ Side effects

- ❑ Nausea and vomiting –most common
- ❑ Edema , hyper pigmentation
- ❑ Hypertension and breast tenderness
- ❑ Cervical or vaginal adenocarcinoma among the daughters of women who took the drug during early pregnancy. E.g. diethylstilbestrol
- ❑ Post-menopausal uterine bleeding
- ❑ Small increase in the incidence of breast cancer
- ❑ Increased risk of endometrial carcinoma if taken alone but concomitant use with progestin prevents this increased risk

➤ Contraindications

- ❑ Estrogen dependent neoplasmas such as carcinoma of the endometrium and breast
 - ❑ Undiagnosed genital bleeding
 - ❑ Liver disease
 - ❑ History of thromboembolic disorder
 - ❑ Heavy smokers
-

➤ Antiestrogens

❑ Clomiphene

- Has antioestrogenic effects on breast and uterus but oestrogenic effects on bone, lipid metabolism and blood coagulation
- use-infertility

❑ Tamoxifen

- Has antioestrogenic action on mammary tissue but oestrogenic actions on plasma lipids
- Use-palliative treatment of advanced breast cancer in postmenopausal women

❑ Raloxifen (selective estrogen modulator)

- Use-approved only for the preventive treatment of osteoporosis in postmenopausal women

Progensins

➤ Includes

1. Progesterone and derivatives
 - Progesterone
 - Hydroxyprogesterone
 - Medroxyprogesterone
2. 17-ethyl testosterone derivatives
 - Dimethisterone
3. 19-nortestosterone derivatives
 - Desogestrel (no androgenic activity)
 - Lynestrenol
 - Norethynodrel
 - Norgestimate

- Norethindrone
- L-norgestrel
- Ethynodiol
- Gestodene

➤ Progestins with out androgenic activity include desogestrel,norgestimate and gestodene

➤ Pharmacological action

- ❑ Increased body temperature
 - ❑ Responsible for alveolar development of the secretory apparatus in the breast
 - ❑ Cause the maturation and secretory changes in the endometrium that are seen following ovulation
-

➤ Therapeutic uses

- ❑ Contraception
- ❑ Hormone replacement therapy
- ❑ Secondary amenorrhea, uterine bleeding disorders, luteal-phase support to treat infertility and premature labor
- ❑ Palliative measure for metastatic endometrial carcinoma and
- ❑ Megestrol acetate is used as a second line treatment for breast cancer

➤ Side effects

- ❑ Edema
 - ❑ Increased BP
 - ❑ Depression
 - ❑ The most androgenic progestins are also reduce plasma HDL levels in women-increase the ratio of LDL to HDL cholesterol
 - ❑ Thrombophlebitis
 - ❑ Pulmonary embolism
 - ❑ Acne
 - ❑ Hirsutism and weight gain
-

➤ Antiprogestins

➤ Mefepristone

➤ Use

- ❑ In combination with misoprostol for the termination of early pregnancy
 - ❑ Postcoital contraceptives; the mechanism of action in this case is thought to be prevention of implantation
-

Hormonal contraception

➤ Includes

- ❑ Oral
- ❑ Parenteral and
- ❑ Implanted contraception

■ Oral contraceptives

1. The combined pill (estrogen with progestin)
2. Progestin only pill

➤ The combined pill

- ❑ Most commonly used
 - ❑ Estrogen- ethinylestradiol , mestranol
 - ❑ Progestin- norethindrone , levonorgestrel , desogestrel
 - ❑ The combined pill is taken for 21 consecutive days followed by 7 pill-free days
-

➤ Mechanism of action

- ❑ Estrogen inhibits the release of FSH and thus suppresses the development of the ovarian follicle
 - ❑ Progestin inhibits the release of LH and prevent ovulation, it also makes the cervical mucus less suitable for the passage of sperm
 - ❑ Together they alter the endometrium in such a way as to discourage implantation
 - ❑ They may also interfere with the coordinated contractions of cervix, uterus and fallopian tubes, which are thought to be necessary for successful fertilization and implantation
-

➤ Preparations

- ❑ 20 – 50 mcg(microgram) estrogen
- ❑ 0.15 – 1 mg progesterone in each tablet
- ❑ Eg.microgynon
 - Ethylestradiol – 30mcg
 - Levonorgestrel -0.15 mg
- ❑ One tablet daily for 21 days starting on fifth day of menstruation , then 7 tablet free days or sometimes it has 7 tab of iron preparation or
 - Ethinylestradiol – 35 mcg
 - Norethindrone - 50 mcg

➤ Side effects

- ❑ Weight gain, nausea, mood changes
 - ❑ Breast fullness, skin pigmentation
 - ❑ Edema, headache
 - ❑ Reversible hypertension
 - ❑ Increased risk of breast cancer-controversial
 - ❑ Increased risk of thromboembolism
-

➤ Contraindications

- ❑ Cerebrovascular and thromboembolic disease
- ❑ Estrogen-dependent neoplasms (e.g. genital tract malignancies)
- ❑ Liver disease
- ❑ Migrane

➤ Drug interactions

- ❑ Anticonvalsants ,hypnotics ,
antibiotics ,antacids,anticoagulants,antidepressants ,steroids and hypoglycemic agents

➤ Progestin-only pill

- ❑ The drugs used include norethindrone, levonorgestrel, ethynodiol
- ❑ The pill is taken daily without interruption
- ❑ Are less effective than the combination pill

➤ Mechanism

- ❑ Primarily on the cervical mucus which is made inhospitable to sperm
- ❑ Also hinders implantation through its effect on the endometrium and on the motility and secretion of the fallopian tube

-
- An advantage of progestin-only pill can be taken after parturition, as unlike estrogen – containing pills, it does not interfere with lactation
 - Indicated in case of estrogen contraindication as in lactating mothers, diabetics, hypertensive, hepatic disease, psychosis or mental retardation or prior thromboembolism
 - Preparation
 - Lynestrenol 0.5mg/day
-

➤ Side effects

- ❑ Irregular vaginal bleeding-most common
 - ❑ Headache
 - ❑ Mood change
 - ❑ Dizziness
 - ❑ Weight gain
 - ❑ Acne
 - ❑ Hirsutism
-

Others

- *Postcoital oral contraceptives/emergency contraception/morning after pill*
 - ❑ Contraceptives aimed at preventing pregnancy after unplanned sexual exposure in women who is not on regular contraception
 - ❑ When treatment began within 72 hrs, it is effective 99% of the time

-
- Combined oral contraceptive (COC) with 50 microgram of estrogen ,2 tabs 12 hrly or
 - COC with 35 microgram of estrogen 4 tabs 12hrly within 72 hrs of unplanned sexual exposure for two dose or
-

➤ Side effects

- ❑ Nausea or vomiting-40%-administered with antiemetics
 - ❑ Headache
 - ❑ Dizziness
 - ❑ Breast tenderness
 - ❑ Abdominal and leg cramp
-

■ Injectables

- ❑ Medroxyprogesterone acetate: 150 mg deep IM injection with in the first 5 days of the cycle to be repeated every 3 months
- ❑ After a 150 mg dose, ovulation is inhibited for at least 14 weeks

■ Side effects

- ❑ Menstrual irregularities are common
- ❑ Infertility may persist for many months after cessation of treatment
- ❑ Almost all users experience episodes of unpredictable spotting and bleeding ,particularly during the first year of use
- ❑ Amenorrhea is common

➤ Implants

- ❑ Levonorgesterel (norplant): in six silastic capsules implanted in the left upper arm under local anesthesia, effective up to five years
- ❑ Disadvantage: need for surgical insertion and removal of capsules

➤ Side effects

- ❑ Irregular menstrual bleeding
- ❑ Headache or visual disturbance
- ❑ A levonorgestrel –impregnated intrauterine device : has contraceptive action for 3-5 years

Androgens and antiandrogens

➤ Androgens include

- ❑ Testosterone cypionate
- ❑ Danazol
- ❑ Fluoxymesterone
- ❑ Nandrone
- ❑ Stanozolol
- ❑ Methandrostenolone (methandienone)
- ❑ Oxymetholone
- ❑ Ethylestrenol
- ❑ Oxandrolone
- ❑ Dromostanolone

➤ Therapeutic uses

- ❑ Replacement therapy in testicular failure
 - ❑ As anabolic agents
 - ❑ Endometriosis (ectopic growth of the endometrium)-
danazol, a mild androgen
-

➤ Andtiandrogens

- ❑ Cyproterone acetate
- ❑ Finasteride
- ❑ Flutamide

➤ Therapeutic uses

- ❑ Used as part of the treatment of the prostatic cancer-
flutamide,cyproterone
- ❑ To treat hirsutism in females –cyproterone
- ❑ Benign prostatic hypertrophy - finasteride

Adrenocorticosteroids and adrenocortical antagonists

- The adrenal cortex releases a large number of steroids into the circulation
- The hormonal steroids may be classified into
 1. Those having important effects on intermediary metabolism
 - Glucocorticoids: hydrocortisone (cortisol), corticosterone
 2. Those having principally salt-retaining activity
 - Mineralocorticoids : aldosterone
 3. Those having androgenic or estrogenic activity.

Corticosteroids

➤ Glucocorticoids

1. Short acting 8-12 hrs
 - hydrocortisone (cortisol), corticosterone
2. Intermediate acting 18-36 hrs
 - Prednisone, prednisolone, methylprednisolone, triamcinolone, meprednisone, fluprednisolone
3. Long acting 1-3 days
 - Betamethasone, paramethasone, dexamethasone, beclomethasone

➤ Mineralocorticoids

- Fludrocortisone, deoxycortisone : used with glucocorticoid in replacement therapy

Compound	Relative affinity for glucocorticoid receptors ^a	Approximate relative potency in clinical use		Duration of action after oral dose ^b	Comments
		<i>Anti-inflammatory</i>	<i>Sodium retaining</i>		
Hydrocortisone ^{Rx} (cortisol)	1	1	1	Short	Drug of choice for replacement therapy
Cortisone	0.01	0.8	0.8	Short	Cheap; inactive until converted to hydrocortisone; not used as anti-inflammatory because of mineralocorticoid effects
Corticosterone	0.85	0.3	15	Short	-
Prednisolone ^{Rx}	2.2	4	0.8	Intermediate	Drug of choice for systemic anti-inflammatory and immunosuppressive effects
Prednisone ^{Rx}	0.05	4	0.8	Intermediate	Inactive until converted to prednisolone ^{Rx}
Methylprednisolone ^{Rx}	11.9	5	Minimal	Intermediate	Anti-inflammatory and immunosuppressive
Triamcinolone ^{Rx}	1.9	5	None	Intermediate	Relatively more toxic than others
Dexamethasone ^{Rx}	7.1	30	Minimal	Long	Anti-inflammatory and immunosuppressive, used especially where water retention is undesirable (e.g. cerebral oedema); drug of choice for suppression of adrenocorticotrophic hormone production
Betamethasone ^{Rx}	5.4	30	Negligible	Long	Anti-inflammatory and immunosuppressive, used especially when water retention is undesirable
Deoxycortone	0.19	Negligible	50	-	-
Fludrocortisone	3.5	15	150	Short	Drug of choice for mineralocorticoid effects
Aldosterone	0.38	None	500	-	Endogenous mineralocorticoid

Glucocorticoids

➤ Pharmacological actions

- ❑ Promote normal intermediary metabolism (i.e. of carbohydrates, proteins, fat)
- ❑ Increase resistance to stress
- ❑ Alter blood cell levels in plasma
- ❑ Anti-inflammatory and immunosuppressive action
- ❑ Reduced vasodilatation, decreased fluid exudation
- ❑ High dose stimulate gastric acid and pepsin production and may exacerbate ulcer

➤ Therapeutic uses

- ❑ Replacement therapy for patients with adrenal failure .E.g. Addison's disease
- ❑ Anti inflammatory/immunosuppressive therapy
 - Asthma
 - Topically in various inflammatory conditions of skin,eye,ear, or nose (e.g. eczema,allergic conjunctivitis,rhinitis)
 - Hypersensitivity states (e.g. severe allergic reactions to drugs or insect venom)
 - Disease with autoimmune and inflammatory components . e.g rheumatoid arthritis and other connective tissue disease, inflammatory bowel disease,some forms of haemolytic anemia
 - To prevent graft – verses- host disease following organ or bonemarrow transplantation

❑ Neoplastic disease

- In combination with cytotoxic drugs in treatment of specific malignances. E.g Hodgkin's disease, acute lymphocytic leukemia
 - To reduce cerebral edema in patient with metastatic or primary brain tumors-dexamethasone
 - As components of anti-emetic treatment in conjunction with chemotherapy
-

➤ Side effects

- ❑ Negative calcium balance, osteoporosis
 - ❑ Suppression of the response to infection or injury
 - ❑ Suppression of a patient's capacity to synthesize corticosteroids-withdraw gradually in prolonged therapy
 - ❑ Increased ICP, muscle wasting
 - ❑ Glaucoma
 - ❑ Hypercoagulability of the blood, disorders of menstruation
 - ❑ Increased appetite, hypertension, peptic ulcers, edema
-

-
- Inhibitors of adrenocorticoid (glucocorticoid) biosynthesis
 - ❑ Metyrapone, ketoconazole-used for treatment of Cushing's syndrome
 - ❑ Aminogluthimide- used for treatment of breast cancer, malignancies of adrenal cortex
 - Mineralocorticoid receptor antagonists
 - ❑ Spironolactone – hyperaldosteronism-potassium sparing
-